

Literature Status Note: the Sub-Markovian Littlewood–Paley–Stein Question

Run `fa_banach_001`, lane 7

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Status. Literature already answered. The source question is from Teresa Martínez, José L. Torrea, and Quanhua Xu, *Vector-valued Littlewood-Paley-Stein theory for semigroups*, arXiv:math/0505303, Adv. Math. 203 (2006), 430–475. The answering paper is Quanhua Xu, *Holomorphic functional calculus and vector-valued Littlewood-Paley-Stein theory for semigroups*, arXiv:2105.12175.

The source question. In Section 3 of arXiv:math/0505303, Theorems 3.1 and 3.2 (**lusin semigroups** and **lusin type semigroups**) characterize martingale cotype and martingale type by vector-valued Littlewood–Paley–Stein inequalities for the Poisson semigroup $\{P_t\}_{t>0}$ subordinated to every symmetric diffusion semigroup $\{T_t\}_{t>0}$. Immediately after the duality remarks following these theorems, the authors write that all semigroups considered in that paper are Markovian ($T_t 1 = 1$), and that they do not know whether Theorems 3.1 and 3.2 still hold for symmetric sub-Markovian semigroups, meaning the same assumptions except Markovianity.

The answering theorem. Xu’s 2021 paper explicitly identifies this as an open problem from the earlier Littlewood–Paley–Stein papers. In the introduction it says that Rota’s dilation prevented weakening the symmetric diffusion hypothesis, that Cowling’s approach applies to symmetric sub-markovian semigroups, and that it had been open to establish the results of [1] in Cowling’s setting. The paper then states that its objective is to resolve those problems.

The decisive result is Theorem 1.1, labelled **Poisson ML** in arXiv:2105.12175.

Let X be a Banach space and $1 < p, q < \infty$. Let $\{T_t\}_{t>0}$ be a strongly continuous semigroup of regular operators on $L_p(\Omega)$ and let $\{P_t\}_{t>0}$ be its subordinated Poisson semigroup. If X has martingale cotype q , then X has Luzin cotype q relative to $\{P_t\}_{t>0}$. If X has martingale type q , then X has Luzin type q relative to $\{P_t\}_{t>0}$.

The same introduction notes immediately after the theorem that this improves Theorem A from the earlier work because the semigroup is only assumed on a single $L_p(\Omega)$, and markovianity, submarkovianity, and symmetry are not assumed.

Why this answers the source question. A symmetric sub-Markovian semigroup is, in particular, a positive contraction semigroup on the relevant L_p spaces, and hence a semigroup of regular operators on each fixed $L_p(\Omega)$. Applying Xu’s Theorem **Poisson ML** to such a semigroup gives the cotype and type inequalities for its subordinated Poisson semigroup. Thus the “if” directions requested in the sub-Markovian extension of Theorems 3.1 and 3.2 hold in a strictly broader setting.

For the converse directions in the original “for every semigroup” formulation, no new argument is needed: Markovian symmetric diffusion semigroups are a subclass of symmetric sub-Markovian semigroups, so the original 2006 theorem already forces martingale cotype/type when the inequality is required for every symmetric sub-Markovian semigroup.

Scope. This note records the literature answer to the Poisson-subordinate sub-Markovian extension of Theorems 3.1 and 3.2 from arXiv:math/0505303. It is not a new proof or counterexample from this run. It does not settle the source paper’s separate Problem 1 on weak type $(1, 1)$, nor all non-subordinated heat-semigroup variants outside the hypotheses of Xu’s later theorems.

References

- [1] T. Martínez, J. L. Torrea, and Q. Xu, *Vector-valued Littlewood-Paley-Stein theory for semigroups*, Adv. Math. 203 (2006), 430–475; arXiv:math/0505303.
- [2] Q. Xu, *Holomorphic functional calculus and vector-valued Littlewood-Paley-Stein theory for semigroups*, arXiv:2105.12175.